```python
from collections import deque

def water_jug(cap1, cap2, target):
    queue = deque([(0, 0, [])])
    visited = set()

    while queue:
        a, b, path = queue.popleft()

        if (a, b) in visited:
            continue

        visited.add((a, b))
        path = path + [(a, b)]

        if a == target or b == target:
            for step in path:
                print(step)
            print("Target reached:", target, "litres")
            return

        states = [
            (cap1, b),
            (a, cap2),
            (0, b),
            (a, 0),
            (a - min(a, cap2 - b), b + min(a, cap2 - b)),
            (a + min(b, cap1 - a), b - min(b, cap1 - a))
        ]

        for state in states:
            if state not in visited:
                queue.append((state[0], state[1], path))

water_jug(4, 3, 2)
```

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/AIES.py ==========
========
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
Target reached: 2 litres
>>>
```